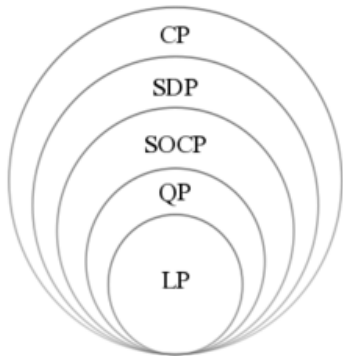


Nonlinear Programs Solvers



- Sequential quadratic programming
- Penalty methods
- Barrier / interior-point methods
- Constrained optimization in R



SEQUENTIAL QUADRATIC PROGRAMMING

- For simplification, consider only equality constraints:

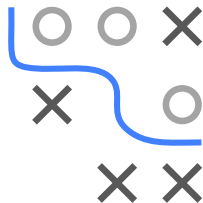
$$\min f(\mathbf{x}) \quad \text{s.t.} \quad h(\mathbf{x}) = 0$$

Idea: Instead of f , optimize 2nd order Taylor approximation at point $\tilde{\mathbf{x}}$

$$\tilde{f}(\mathbf{x}) = f(\tilde{\mathbf{x}}) + \nabla_{\mathbf{x}} f(\tilde{\mathbf{x}})^T (\mathbf{x} - \tilde{\mathbf{x}}) + \frac{1}{2} (\mathbf{x} - \tilde{\mathbf{x}})^T \nabla_{\mathbf{xx}}^2 f(\tilde{\mathbf{x}}) (\mathbf{x} - \tilde{\mathbf{x}})$$

- h is also replaced by its linear approximation at $\tilde{\mathbf{x}}$

$$\tilde{h}(\mathbf{x}) = h(\tilde{\mathbf{x}}) + \nabla h(\tilde{\mathbf{x}})^T (\mathbf{x} - \tilde{\mathbf{x}})$$

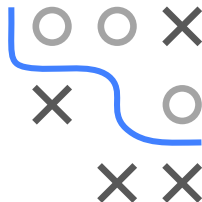


SEQUENTIAL QUADRATIC PROGRAMMING

- With $\mathbf{d} := (\mathbf{x} - \tilde{\mathbf{x}})$ we formulate the **quadratic auxiliary problem**

$$\begin{aligned} \min_{\mathbf{d}} \quad & \tilde{f}(\mathbf{d}) := f(\tilde{\mathbf{x}}) + \mathbf{d}^T \nabla_{\mathbf{x}} f(\tilde{\mathbf{x}}) + \frac{1}{2} \mathbf{d}^T \nabla_{\mathbf{xx}}^2 f(\tilde{\mathbf{x}}) \mathbf{d} \\ \text{s.t.} \quad & \tilde{h}(\mathbf{d}) := h(\tilde{\mathbf{x}}) + \nabla h(\tilde{\mathbf{x}})^T \mathbf{d} = 0 \end{aligned}$$

- Even if no optimality conditions can be formulated for the actual problem, KKT conditions apply at the optimum of this problem
- If $\nabla_{\mathbf{xx}}^2 f(\mathbf{x})$ is positive semidefinite, it is a **convex optimization problem**



SEQUENTIAL QUADRATIC PROGRAMMING

- Using the Lagrange function

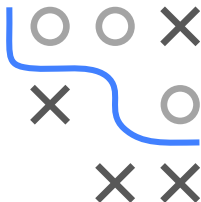
$$L(\mathbf{d}, \beta) = \mathbf{d}^T \nabla_x f(\tilde{\mathbf{x}}) + \frac{1}{2} \mathbf{d}^T \nabla_{xx}^2 f(\tilde{\mathbf{x}}) \mathbf{d} + \beta^T (h(\tilde{\mathbf{x}}) + \nabla h(\tilde{\mathbf{x}})^T \mathbf{d})$$

we formulate the KKT conditions

- $\nabla_{\mathbf{d}} L(\mathbf{d}, \beta) = \nabla_x f(\tilde{\mathbf{x}}) + \nabla_{xx}^2 f(\tilde{\mathbf{x}}) \mathbf{d} + \nabla h(\tilde{\mathbf{x}})^T \beta = 0$
- $h(\tilde{\mathbf{x}}) + \nabla h(\tilde{\mathbf{x}})^T \mathbf{d} = 0$
- Or in matrix notation

$$\begin{pmatrix} \nabla_{xx}^2 f(\tilde{\mathbf{x}}) & \nabla h(\tilde{\mathbf{x}})^T \\ \nabla h(\tilde{\mathbf{x}}) & 0 \end{pmatrix} \begin{pmatrix} \mathbf{d} \\ \beta \end{pmatrix} = - \begin{pmatrix} \nabla_x f(\tilde{\mathbf{x}}) \\ h(\tilde{\mathbf{x}}) \end{pmatrix}$$

- The **quadratic subproblem** can be traced back to solving a linear system



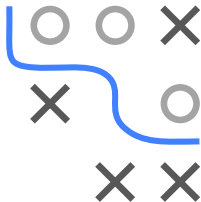
SQP ALGORITHM

Algorithm SQP for problems with equality constraints

- 1: Select a feasible starting point $\mathbf{x}^{(0)} \in \mathbb{R}^n$
- 2: **while** Stop criterion not fulfilled **do**
- 3: Solve quadratic subproblem by solving the equation

$$\begin{pmatrix} \nabla_{xx}^2 L(\mathbf{x}, \mu) & \nabla h(\mathbf{x})^T \\ \nabla h(\mathbf{x}) & 0 \end{pmatrix} \begin{pmatrix} \mathbf{d} \\ \beta \end{pmatrix} = - \begin{pmatrix} \nabla_x L(\mathbf{x}, \mu) \\ h(\mathbf{x}) \end{pmatrix}$$

- 4: Set $\mathbf{x}^{(i+1)} = \mathbf{x}^{(i)} + \mathbf{d}$
 - 5: **end while**
-



PENALTY METHODS

Idea: Replace constrained problem with a sequence of unconstrained problems using a **penalty function**

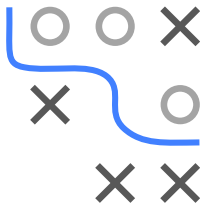
- Instead of

$$\min_{\mathbf{x}} f(\mathbf{x}) \quad \text{s.t.} \quad h(\mathbf{x}) = 0$$

we look at the unconstrained problem

$$\min_{\mathbf{x}} p(\mathbf{x}) = f(\mathbf{x}) + \rho \frac{\|h(\mathbf{x})\|^2}{2}$$

- Under appropriate conditions, solutions for $\rho \rightarrow \infty$ converge to the solution of the original problem



BARRIER METHOD

Idea: Establish a “barrier” that penalizes if $\mathbf{x}$ comes too close to the edge of the feasible set $\mathbf{S}$

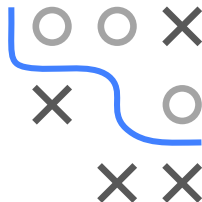
- For the problem

$$\min f(\mathbf{x}) \quad \text{s.t.} \quad g(\mathbf{x}) \leq 0$$

a common **barrier function** is

$$B_\rho = f(\mathbf{x}) - \rho \sum_{i=1}^m \ln(-g_i(\mathbf{x}))$$

- Penalty term becomes larger as $\mathbf{x}$ approaches 0, i.e., the limit of the feasible set
- Under certain conditions, solutions of $\min B_\rho$ for $\rho \rightarrow 0$ converge to the optimum
- Also called **interior-point method**



CONSTRAINED OPTIMIZATION IN R

- `optim(..., method = "L-BFGS-B")` uses quasi-Newton methods and can handle box constraints
- `nlminb()` uses trust-region procedures and can also handle box constraints
- `constrOptim()` can be used for linear inequality constraints and is based on interior-point methods
- `nloptr` is an interface to NLOpt, an open-source library for nonlinear optimization

